



Management of mango fruit fly, *Bactrocera dorsalis*

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Abstract

Mango, India’s second most vital fruit crop, is dubbed the "King of fruits" for its exquisite flavor, rich nutrients and global economic value. However, fruit flies, among other pests, severely impair its yield and quality, rendering infested fruits unsellable and inedible. While insecticides work, they leave residues that hinder international trade. To address this, sustainable and safe methods are essential for farmers to control fruit flies locally and ensure market-ready produce. Integrated pest management is thus recommended, combining various eco-friendly tactics to reduce pest populations, while safeguarding beneficial insects and preserving mango quality for domestic and global markets.

Introduction

Mangoes are a seasonal crop, and insect problems vary throughout the year. One of the most destructive pests is the *Bactrocera* genus of fruit flies, particularly in tropical and subtropical regions like south India. These flies damage a variety of soft fruits and vegetables, including mangoes. The most common species affecting mangoes are *Bactrocera dorsalis* (oriental fruit fly), *Bactrocera zonata* (peach fruit fly), and *Bactrocera correcta* (guava fruit fly). Fruit flies can cause significant yield losses, with 35-40% of mangoes damaged during harvest and, in some cases, up to 80% of the yield lost. This infestation is a key factor in reducing mango exports and is considered a major quarantine pest in India.

Taxonomic classification

Mango fruit fly, belongs to the Domain: Eukaryota, Kingdom: Metazoa, Phylum: Arthropoda, Subphylum: Uniramia, Class: Insecta, Order: Diptera, Family: Tephritidae, Genus: *Bactrocera* and Species: *dorsalis*.

Life cycle

As a holometabolous insect the life cycle of mango fruit fly have four distinct stages: egg, larva (maggot), pupa and adult.



Symptoms of damage

Initial Signs:

- Punctures, sunken patches, softening, discoloration, oozing and foul odour.

Advanced Stages:

- **Maggots:** Once the eggs hatch, the larvae (maggots) feed on the fruit pulp, causing it to decay and rot.
- **Significant Softening:** The fruit becomes mushy and may wrinkle.
- **Fruit Drop:** Infested fruits may drop prematurely from the tree.
- **Foul Smell:** The affected fruit emits a strong, foul smell.
- The fruits may discolour and show stains of mold as they progressively decay



IPM Strategies

Fruit flies are difficult to handle due to their high polyphagia, fast reproduction, and regional adaptation. Around the world, chemical pesticides are utilised to control the fruit fly threat. However, chemical pesticides are often associated with serious health and environmental risks. Mango fruits are among the horticulture products that have been snatched from global markets as a result of the overuse of chemical pesticides. Furthermore, the majority of farmers with limited resources cannot afford the high expense of pesticides. The widespread application of broad-spectrum conventional insecticides and their replacement by Integrated Pest Management (IPM), which is more ecologically friendly.

Cultural methods

- Regularly collect and destroy fallen fruits, which are breeding grounds for fruit fly larvae.
- Bury infested fruits deep in the ground or use a hammer mill to destroy them.
- Plough the soil around the tree basin to expose pupae to sunlight and predators.
- Harvest mature fruits early to avoid fruit fly infestation.
- Bagging fruits with double-layer brown paper bags can significantly reduce infestation.
- Remove wild guava and other plants that can serve as breeding grounds for fruit flies near the orchard.
- Fruit fly populations can be considerably decreased by crushing fruits in a grinder and then burying them at least 50 cm below the soil's surface (Reddy *et al.*, 2020).

Mechanical Control

- Fruits can be mechanically controlled by wrapping, bagging, or netting them in paper bags or newspaper. In order to produce fruits free of fruit fly infestation, this is done to prevent adult female flies from laying their eggs on the fruits. Fruits should be wrapped at least one month before harvesting, or before fruit flies strike (Reddy *et al.*, 2020).
- Despite being a time-consuming procedure, this works well to stop fruit flies from attacking fruits (Badii *et al.*, 2015).

Physical Control

- Fruit fly control can be achieved physically through post-harvest quarantine measures. For the export commodities, these treatments are required. Post-harvest quarantine treatments that are available and being utilised in place of harmful chemicals. The following are examples of fumigation: i) heat treatment to raise fruit temperatures over fruit fly thermal limits and ii) cold treatment to lower fruit temperatures below fruit fly thermal limits (Reddy *et al.*, 2020).

Behavioural control

- Three bait trap combinations *viz.*, Methyl eugenol (ME)+Malation, Banana+yeast+spinosad and Fishmeal+diazinon were used against fruit flies on mango (Bhagat *et al.*, 2022).
- The primary method of controlling fruit flies is the combination of a lethal agent with food baits, which are hydrolysed proteins or their ammonium analogues. The Bait Application Technique (BAT) is the name of this method. Both male and female fruit flies are drawn in by these lures. Nowadays, commercial baits like GF-120 (Success® Apart), Buminal, Nulure, and SolBait that are premixed with insecticides like Spinosad for immediate use are currently available on the market (Ekesi *et al.*, 2009).
- Arka Bactro+ a plant based kairomone was developed to attract more numbers of male fruit flies of *Bactrocera dorsalis* and *B. correcta* (ICAR- IIHR, 2017).

Biological control

- Natural predators of fruit flies are vital for their control, so it is essential to protect these natural enemies in farms or orchards (Stibick, 2004). Widespread pesticide spraying can be detrimental to these beneficial organisms. To safeguard them, it is better to reduce blanket pesticide use and apply treatments only where necessary. Relying on natural enemies is a safer, more sustainable, and economical strategy (Ekesi and Billah, 2006).
- *Opines* spp. is the most prevalent group of parasitoids commonly used to control fruit flies. These larval parasitoids target the young larvae of fruit flies that feed beneath the skin of the fruit. Additionally, larval-pupal parasitoids such as *Terastichus* spp. (Eulophids) can be utilized to enhance the effectiveness of *Opines* spp. (Stibick, 2004).
- Entomopathogenic fungi like *Beauveria bassiana* (Balsamo), *Metarhizium anisopliae*, and *Isaria fumosorosea* (Wize) caused a 91% mortality rate in *Rhagoletis cerasi* (L.) (Daniel and Wyss, 2009). Entomopathogenic nematodes such as *Steinernema carpocapsae* Weiser (formerly *Neoaplectana carpocapsae*) have shown to significantly reduce melon fruit fly populations under laboratory conditions (Ganie *et al.*, 2022).

Chemical control

- Use of new molecules: Spray of Lambda-cyhalothrin 5% EC @ 2 ml/litre (Karate®) or thiamethoxam (12.6%) + Lambda-cyhalothrin (9.5%) ZC (Alika® @ 0.4 ml/ litre or Quinalphos 25% EC (Ekalux® @ 2 ml/ litre + neem oil (Econem Plus @ 0.3%) will help in controlling fruit flies under their high population situation in mango (Kumar Singh *et al.*, 2022).

Conclusion

Fruit fly infestations in mangoes can cause up to 90% damage. Integrated pest management strategies include methyl-eugenol traps, neem-based sprays, protein bait sprays, male annihilation techniques, insecticidal cover sprays, orchard sanitation, and other practices to reduce damage. These methods help control fruit fly populations during fruit maturity.

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CATALOGUES